

Brandon Liu

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Education

Purdue University

B.S in Mechanical Engineering, Minor in Statistics

West Lafayette, IN

Aug 2023 - May 2027

Professional Experience

agrobotics

Mechanical Engineer Intern

Salinas, CA

Jan 2026 - Aug 2026

- Led the design, build of a 7 DoF humanoid arm using Fusion 360, resulting in a 55% reduction in weight and a 63% increase in manipulability compared to the previous arm.
- Led the end to end design cycle of new drive units using design for manufacturability and assembly techniques resulting in a 40% manufacturing cost reduction and a 250% increase in in-field chassis stability.
- Designed and built 6 low cost data-collection units , achieving a 90% uptime in 16/7 shifts while being used in agricultural fields.
- Assisted in the bring up of robotic arms and drive units through the tuning of low-level PID position and velocity controllers, resulting in a reduction of steady state error.
- Created design requirements and benchmarked compute solutions resulting in a reduction of component count and ruggedization of the electrical subsystems.
- Created and tested CAN and RS485 harnesses for actuators and end effectors.

Purdue ACM SIGBots

Lead Design Engineer

West Lafayette, IN

Aug 2023 - Present

- Led an interdisciplinary team of 60+ students in the end-to-end design, fabrication, and documentation of 6 competitive robotics platforms.
- Established and managed onboarding and training pipelines for 20+ new members annually, advancing team-wide proficiency in CAD and manufacturing tools.
- Secured \$15,000 in sponsorship and event revenue by coordinating large-scale robotics competitions and industry outreach.

ASME Energy, Fuel Cell

Mechanical Engineer

West Lafayette, IN

Aug 2025 - Present

- Perform design review sessions of the fuel cell and secured \$5,000 in funding through detailed technical justifications and project presentations.
- Design evaluation of a 200W, 9V output, hydrogen fuel cell system in Fusion 360.
- Contribute to early fabrication planning, ensuring component tolerances and manufacturability align with design requirements.

ACS System Associates

Project Engineer Intern

Mount Vernon, NY

Jun 2025 - Aug 2025

- Coordinated with vendors and subcontractors to generate precise cost estimates for both new and retrofit HVAC projects.
- Oversaw delivery schedules and inventory systems for instrumentation, streamlining jobsite logistics and reducing material delays.

Semiconductor Packaging Laboratory, Purdue University

Undergraduate Research Fellow, Instructor: Dr. Tiwei Wei

West Lafayette, IN

Aug 2024 - Dec 2024

- Developed a MATLAB-based 3D fluid topology optimization algorithm using the Method of Moving Asymptotes (MMA), improving manifold thermal efficiency and reducing pressure drop by 5%.
- Designed and fabricated novel manifold prototypes via SLA 3D printing and SolidWorks for presentation at CES 2025.

Projects

Competitive Robotics [Robot Portfolio]

Aug 2023 - Present

Awards: 2x World Champion, 5x World Championship Awards, 20+ Regional Awards

- Owned end to end design process for 11 robots. Team lead in CAD, fabrication, testing and conducted weekly design reviews
- Designed, fabricated, and implemented custom differential lifts, drive chassis, end effectors, [more...]
- Created custom curriculum to train 50+ members with in CAD, fabrication, CNC manufacturing, 3DP, laser cutting. Increasing training and member proficiency by 40%

Novel Step Stool

Jan 2025 - May 2025

- Collaborated with a four-member team to design a high-fidelity, portable step stool prototype in Autodesk Inventor, enhancing portability and storage efficiency.
- Engineered and fabricated a full-size prototype using FDM 3D Printing and CO₂ Laser Cutting after resolving stability and tipping issue with initial scaled-down prototype.

Technical Skills

CAD & Analysis: SolidWorks, NX, Inventor, Simulink, Fusion 360, Onshape, LTSpice

Manufacturing: Design for Manufacturing (DFM), GD&T, 3D Printing, CNC Routing, Laser Cutting, Molding

Programming: MATLAB, R, C, Python, Java, C++